Campus Companion

Software Requirements Specification

Contents

1 Introduction **1**

2 Users **1**

3 System overview **1**

4 Functional requirements **1**

5 Data requirements **3**

6 Non-functional requirements **3**

7 Main screens **3**

8 Testing and acceptance **3**

9 Conclusion **4**

1 Introduction

1.1 Purpose

This document describes the first version of Campus Companion. It tells the team what must be implemented and gives simple checks for the final demonstration.

1.2 Product description

Campus Companion is a mobile application for Android and iOS. It helps students find buildings, view campus information and follow walking routes on a simple 3D map. It also provides announcements, events and campus services.

1.3 Project limits

The project covers one university campus. The first version will not include payments, chat, indoor positioning, official grades, AI features or QR access control.

2 Users

User	What the user can do
Student or visitor	Search for places, view the map, read information and follow routes.
Content manager	Add or update announcements, events, buildings and services.
Administrator	Manage accounts and give content-management access.

3 System overview

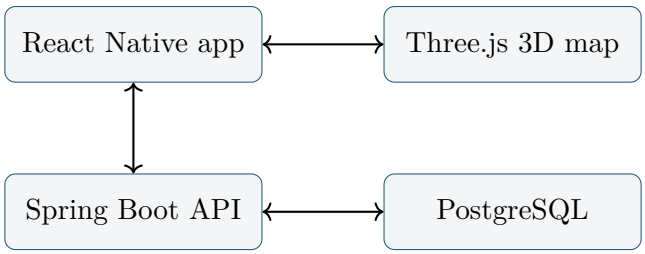


Figure 1: Main parts of the system.

The mobile app contains the normal screens and opens the 3D map. The Java backend sends data to the app and saves changes in PostgreSQL. The administration page uses the same backend.

4 Functional requirements

4.1 Account and home

ID	Requirement
FR-01	The user must be able to sign in with a valid account.
FR-02	The user must be able to sign out.
FR-03	The home screen must show recent announcements and upcoming events.
FR-04	The application must show a clear message when no data is available.

4.2 Campus information

ID	Requirement
FR-05	The user must be able to view a list of campus buildings.
FR-06	Each building page must show its name, description and category.
FR-07	The user must be able to view campus services and their contact information.
FR-08	The user must be able to search for a building or service by name.
FR-09	Selecting a search result must open its information page.

4.3 Events and announcements

ID	Requirement
FR-10	The user must be able to read published announcements.
FR-11	The user must be able to view upcoming and past events.
FR-12	An event page must show its title, date, place and description.
FR-13	Selecting an event place must open the matching building or map point.

4.4 3D map and routes

ID	Requirement
FR-14	The application must display a 3D model of the campus.
FR-15	The user must be able to rotate, move and zoom the map.
FR-16	Important buildings and points of interest must be selectable.
FR-17	Selecting a building must show its name and information.
FR-18	The user must be able to choose a starting place and destination.
FR-19	The application must draw a walking route between known route points.
FR-20	The route must show an approximate distance.
FR-21	If the 3D map cannot load, the building list must remain available.

4.5 Content management

ID	Requirement
FR-22	A content manager must be able to add and edit buildings.
FR-23	A content manager must be able to add and edit events and announcements.
FR-24	Only an authorized account may open the administration page.
FR-25	Deleted or unpublished content must not appear in the mobile application.

5 Data requirements

Data	Main information
User	Name, email, role and account status
Building	Name, description, category and map position
Service	Name, description, contact and building
Event	Title, description, date, place and status
Announcement	Title, text, publication date and status
Route point	Position and connections to other route points

6 Non-functional requirements

ID	Requirement
NFR-01	The main screens should load within about three seconds on a normal connection.
NFR-02	The 3D map should remain usable on the team's oldest test phone.
NFR-03	Passwords and authentication tokens must not appear in logs.
NFR-04	Protected operations must be checked by the backend.
NFR-05	The application must use HTTPS when communicating with the backend.
NFR-06	Core information must remain available as text and lists without using the 3D map.

7 Main screens

- sign-in;
- home;
- announcements;
- events and event details;
- buildings and services;
- search;
- 3D campus map;
- route result;
- profile;
- administration page.

8 Testing and acceptance

The project will be considered complete when the team can demonstrate the following:

1. a user signs in and signs out;
2. the home screen displays announcements and events;

3. a user searches for a building and opens its details;
4. the same building can be selected on the 3D map;
5. the application draws a route between two campus locations;
6. an authorized manager adds an event and it appears in the app;
7. the main functions work on one Android phone and one iPhone;
8. no password or authentication token appears in the application logs.

9 Conclusion

This specification defines a small first version that can be completed by a student team. Extra features should be added only after the listed requirements work correctly.